

Fanuc Robot Project Report

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Lab Section: Wednesday - 11:30 AM

Objectives:

- Design and program a robotic system that can automatically identify and sort metal and non-metal objects.
- Gain practical experience programming a robot using the teach pendant and creating structured robot programs.
- Integrate a metal sensor with the robot controller to detect whether an object is metal or non-metal.
- Implement robot input/output (I/O) communication to read sensor signals and control indicator lights.
- Develop a scalable robot program using registers and counters so that additional pick or place positions could be added in the future.
- Demonstrate how industrial robots can be used for automated material handling and sorting applications.

The Idea:

The idea of this project is to design and program an automated robotic system that can identify and separate metal and non-metal objects. Using a FANUC industrial robot, the system simulates a real process where materials must be sorted quickly and accurately. Automation like this is commonly used in manufacturing plants, recycling facilities, and quality control systems to increase efficiency, reduce manual labor, and improve sorting accuracy.

The Product:

The product is a fully automated metal separation system built using a FANUC robot, a metal sensor, and indicator lights.

The system includes:

- A robotic arm that picks up objects from specified positions
- A metal detection sensor
- Sorting locations (metal side and non-metal side)
- Indicator lights to show the classification result
(Red light = Metal detected, White light = Non-metal detected)

This automated setup demonstrates how robots can be used for material sorting and industrial automation tasks.

The Working:

The system works through a programmed sequence of robotic motions and sensor checks.

Step-by-step process:

1. The robot moves to a predefined pick position where an object is placed.
2. The robot picks up the object using its gripper.
3. It then moves the object to the metal detector sensor position.
4. The metal sensor scans the object to determine whether it is metal or non-metal.
5. Based on the sensor signal: If metal is detected, the robot moves to the metal side, places the object there, and activates the red indicator light. If no metal is detected, the robot moves to the non-metal side, places the object there, and activates the white indicator light.
6. After placing the object, the robot returns to the home position and repeats the process for the next item.

This programmed workflow allows the robot to automatically identify, classify, and sort materials, demonstrating the practical application of robotics in industrial automation.

Implementation:

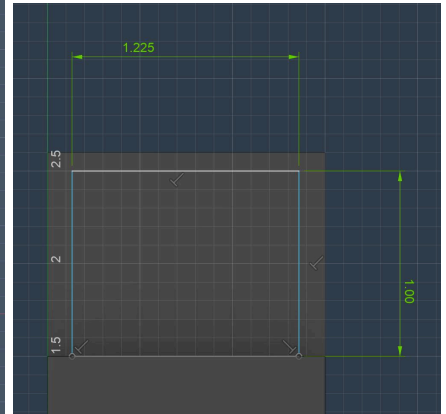
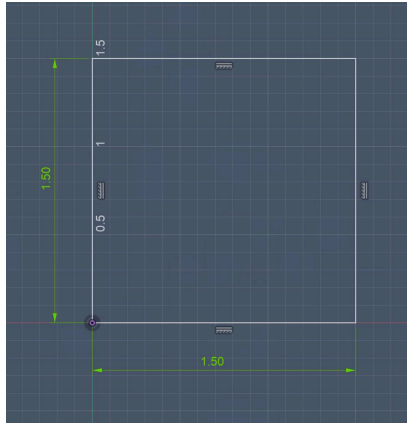
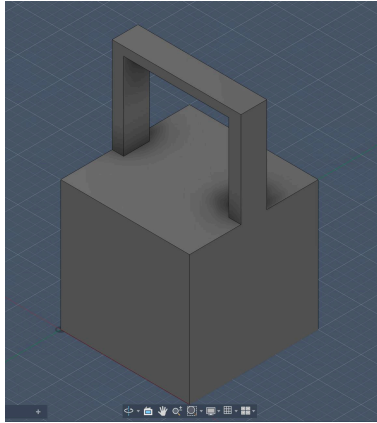
The system was implemented using a combination of custom hardware components and robot programming. Several parts such as the gripper attachment, hook, and sensor stand were designed using CAD and 3D printed for the project. These components allowed the robot to securely pick up the blocks and position them in front of the metal detection sensor for scanning. The proximity sensor was mounted on the sensor stand so that each object could be consistently detected before being sorted. Indicator lights were also connected to the robot outputs to display whether the detected object was metal or non-metal.

The robot program was created using the FANUC teach pendant, and multiple robot positions were stored using position registers (PR). These registers store the robot's coordinates and allow the robot to move to the pick, sensing, and placement locations during the sorting process. A counter register (R[1]) was used in the program to keep track of which object was being processed. Using registers and counters helped organize the program and made it more scalable, since additional pick or sorting positions could easily be added in the future.

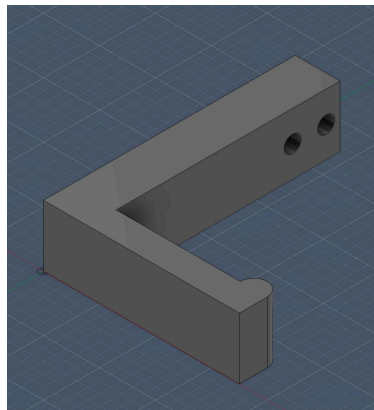
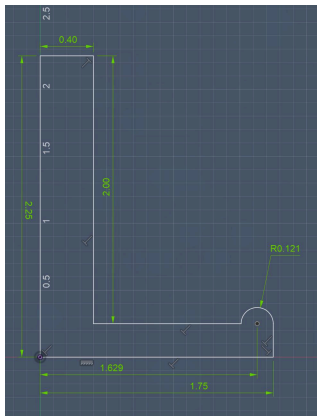
During operation, the robot first moves to the pick position and grabs the object, then moves to the sensing position where the proximity sensor checks whether the object is metal or non-metal. Based on the sensor input, the robot uses conditional logic to move the object to the correct sorting location. If metal is detected, the robot places the object in the metal area and activates the red light. Otherwise, the object is placed in the non-metal area and the white light is activated.

Hardware / Software:

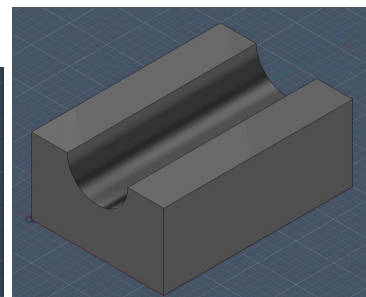
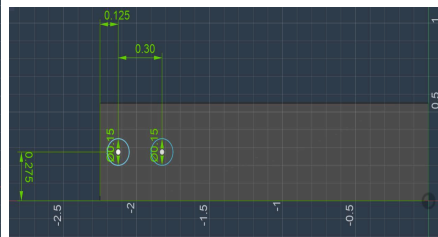
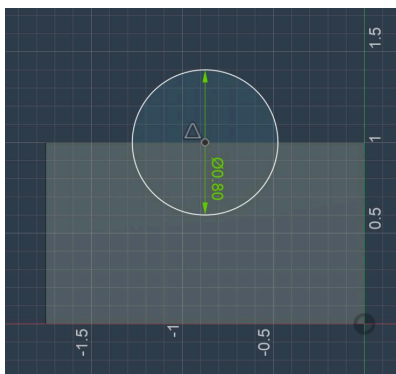
Blocks:



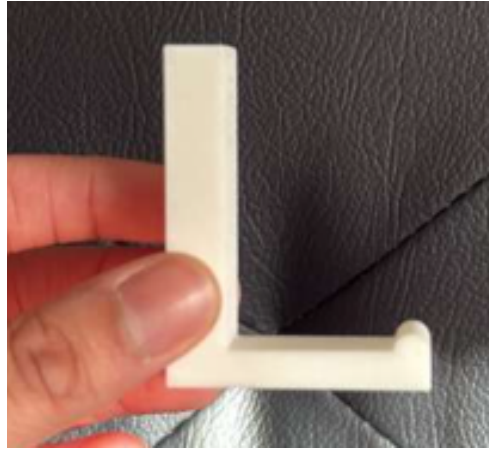
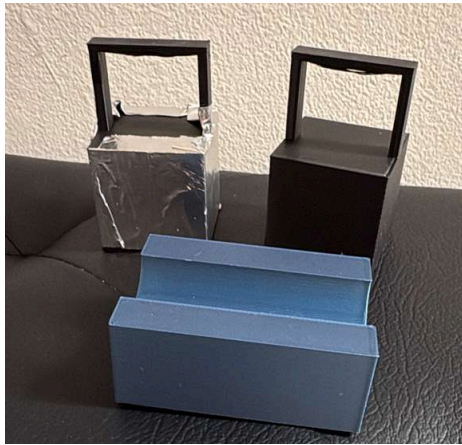
Hook:



Sensor Stand:



3-D Parts:



CODE

```
1: R[1]=0
2: LBL[1]
3: DO[1]=OFF
4: DO[2]=OFF
5: L PR[24] 200mm/sec FINE
6: IF R[1]=0,JMP LBL[5]
7: IF R[1]=1,JMP LBL[6]
8:
9: LBL[5]
10: L PR[20] 200mm/sec FINE
11: L PR[21] 200mm/sec FINE
12: JMP LBL[7]
13: LBL[6]
14: L PR[22] 200mm/sec FINE
15: L PR[23] 200mm/sec FINE
16:
17: LBL[7]
18: L PR[24] 200mm/sec FINE
19: WAIT 2.00(sec)
20: IF DI[3]=ON, JMP LBL[10]
21: DO[1]=ON
22: L PR[26] 200mm/sec FINE
23: L PR[28] 200mm/sec FINE
24: JMP LBL[20]
25: LBL[10]
26: DO[2]=ON
27: L PR[25] 200mm/sec FINE
28: L PR[27] 200mm/sec FINE
29: LBL[20]
30: R[1]=R[1]+1
31: IF R[1]<2,JMP LBL[1]
32: L PR[24] 200mm/sec FINE
33: WAIT 1.00(sec)
34: JMP LBL[1]
35: END
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Challenges Faced:

One challenge we faced during this project was the limited number of position registers available on the robot. Since the robots were shared among many students in the lab, we had to work within the positions that were already available. Initially, our group planned to sort four to six blocks to make the system more complex, but due to the limited position vectors and the need to avoid interfering with other groups' programs, we reduced the design to sorting two blocks. Even though the system handled fewer objects than originally planned, it still allowed us to demonstrate the sensing and sorting logic effectively.

Another challenge involved small issues during programming and testing. While writing and running the robot program, we encountered minor errors such as incorrect jumps between labels or sensor signals not behaving as expected. These issues required several test runs and small adjustments to the code and robot positions. We also faced a small issue during the 3D printing of the gripper component, where the fit needed slight adjustment before it worked properly with the blocks. Through troubleshooting these problems, we were able to refine both the hardware and the program to make the system work reliably.

Learning Outcomes:

Through this project, we gained hands-on experience operating and programming a FANUC robot using the teach pendant. We practiced creating and modifying robot programs directly on the teach pendant, which helped us understand how industrial robots are programmed and controlled.

One of the key skills we developed was teaching and touching up robot positions. We created multiple positions for picking up objects, checking them with the metal sensor, and placing them in the correct sorting area. During testing, we frequently had to touch up points to improve accuracy and ensure the robot moved correctly between positions. We also practiced troubleshooting and debugging robot programs. When the robot did not move as expected or the sensor signals were incorrect, we had to analyze the program and adjust positions, logic, or connections to fix the issue.

Another important concept we practiced was sensor integration and I/O control. The metal sensor provided an input signal to the robot to indicate whether an object was metal or non-metal by turning off or on. Based on this input, the robot program used conditional logic to determine where to place the object. We also used robot outputs to control the indicator lights, turning on the red light for metal objects and the white light for non-metal objects. We used position registers and registers as counters to manage positions and track operations within the program. This helped us understand how registers can be used to organize robot movements and make the program more flexible.

Beyond the technical programming, this project also helped us practice planning and designing a scalable system. Our design allows additional pick positions or sorting locations to be added easily by modifying the counter and adding new positions. The counter and position-based

structure of the program makes it possible to expand the system without completely rewriting the program. This taught us the importance of designing robotic programs that are flexible and expandable rather than limited to a single fixed task.

Industrial Applications:

Automated sorting systems like the one developed in this project are widely used in modern manufacturing and recycling industries. Industrial robots from companies such as FANUC are commonly used to perform repetitive material handling and sorting tasks with high speed and accuracy. In recycling facilities, robots equipped with sensors can automatically separate materials such as metals, plastics, and other recyclable items, improving efficiency and reducing the need for manual labor.

Similar systems are also used in manufacturing and packaging lines, where robots identify and sort products based on material type, size, or other properties. By integrating sensors, robot programming, and automated handling, these systems help increase productivity, improve consistency, and reduce operational costs in industrial environments.

Further Improvements:

Although the system successfully sorts metal and non-metal objects, several improvements could make the project more efficient and reliable.

One possible improvement is optimizing the robot's motion paths and cycle time. Currently, the robot moves between sensing, pick, and place positions sequentially. By refining the path, adjusting speeds, or reducing unnecessary movements, the overall operation could be performed faster and more efficiently.

Another improvement would be to expand the number of pick and sorting positions. Since the program already uses counters and structured logic, additional positions could easily be added. This would allow the system to handle more objects in a single cycle and demonstrate a larger-scale automated sorting process.

The project could also be improved by adding more advanced sensing capabilities. At the moment, the system only distinguishes between metal and non-metal objects. Additional sensors such as vision systems, weight sensors, or size detection could allow the robot to sort objects into more categories, making the system more versatile.

Finally, the program could be improved by adding additional safety and error-handling logic. For example, the robot could check whether an object was successfully picked up before continuing the process, or detect if the sensor fails to provide a signal. Adding these checks would make the system more robust and reliable.